

Unit 12: Weathering & Soils

Reading: pages 164–178 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Which of the following is *not* an agent of mechanical weathering?
 - a) Hydrolysis of feldspar into clay minerals
 - b) The formation of salt crystals within the rock
 - c) The decrease in pressure that follows removal of overlying rock
 - d) Freezing and thawing of water in cracks
2. Frost wedging is most effective in a climate where
 - a) temperatures stay warm year-round
 - b) freezing and thawing alternate frequently
 - c) almost no water is available
 - d) temperatures stay below freezing year-round
3. The chemical weathering of feldspar by hydrolysis produces
 - a) hematite
 - b) clay minerals such as kaolinite
 - c) quartz
 - d) pure calcite
4. Carbonic acid, which drives most chemical weathering, forms when
 - a) water combines with carbon dioxide
 - b) water combines with oxygen
 - c) sulfur dioxide combines with nitrogen
 - d) clay minerals dissolve in rainwater
5. Roughly what share of a typical soil is empty pore space, filled to varying degrees with air and water?
 - a) About 5%
 - b) About 25%
 - c) About 45%
 - d) About 50%

6. Which soil horizon is the eluviated, or leached, layer from which some clay and iron have been removed, leaving it pale and sandier?
- a) O
 - b) A
 - c) E
 - d) B

Part B · True or False

2 questions × 1 point = 2 points

7. Quartz is virtually unaffected by chemical weathering.
TRUE / FALSE
8. Steep slopes tend to develop the thickest, best-developed soils.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Mechanical and chemical weathering are described as helping each other along. Explain how each one assists the other.

10. Name the four factors that control the nature of a soil and the rate at which it forms, and explain the effect of any one of them.

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